Fiscal Year:	FY 2020	**Task Last Updated:**	FY 12/02/2019
PI Name:	Blancaflor, Elison B. Ph.D.		
Project Title:	Leveraging Spaceflight Genomic Data to Uncover Developmental and Cell Type Specific Gene Regulatory Networks in Plants Responding to Gravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential (2) Bioregenerative Life Support		
PI Email:	elison.b.blancaflor@nasa.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	321-501-5368
Organization Name:	NASA Kennedy Space Center		
PI Address 1:	Utilization and Life Science Office		
PI Address 2:	Exploration and Research Technology Programs		
PI Web Page:			
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32955	**Congressional District:**	8
Comments:	NOTE: Dr. Blancaflor moved to NASA Kennedy Space Center in spring 2021; as a PI, his prior grants were for work at the Noble Foundation in Oklahoma.		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	02/01/2019	**End Date:**	07/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:	Ruby, Anna Maria	**Contact Phone:**	321-867-7065
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2021 per transfer of project to Dr. Simon Gilroy, University of Wisconsin (Ed., 9/21/21)		
Key Personnel Changes/Previous PI:	December 2019 report: Dr. Sabrina Chin is a postdoctoral fellow who is supported on this grant. She started on September 9, 2019.		
COI Name (Institution):	Kwon, Taegun Ph.D. (Noble Foundation) Tang, Yuhong Ph.D. (Noble Foundation)		
Grant/Contract No.:	80NSSC19K0129		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	As NASA pursues its goal of sending human colonies to Mars and beyond, there is a pressing need to better understand how the spaceflight environment influences fundamental biological processes. Understanding how plants respond to gravity on Earth and the microgravity environment in space is of particular interest because of the important role that plants will play in advanced bioregenerative life support systems during long duration space missions. In recent years, numerous transcriptomic datasets have shown that hundreds of genes change expression when plants are grown in space. However, it has been challenging to tease apart which genes change due to microgravity from those that change as a result of general plant stress responses. This difficulty stems from the fact that plant spaceflight genomics datasets have been derived almost exclusively from whole seedlings and plant organs, which could potentially mask or dilute gene expression changes that are occurring within the distinct cell-/tissue-types that are responsible for gravity sensing (e.g., root columella and stem endodermal cells). Thus, the major goal of this NASA-ground-based proposal is to begin to take steps to develop methodology that will enable space genomic studies on homogenous cell populations. Here, we propose to use laser capture microdissection (LCM) to conduct RNA-Sequencing (RNA-Seq) on gravity sensing columella cells to identify gene regulatory networks that govern early responses of plants to gravity. We hypothesize that gene profile signatures from a homogenous population of columella cells will more clearly reflect transcriptomic changes that are directly linked to microgravity. The RNA-Seq data that we obtain from our analysis of root columella cells will be compared to columella-specific gene modules we identified through our Weighted Gene Co-expression Network Analysis (WGCNA) of Arabidopsis seedlings grown for 6 and 11 days on the International Space Station (ISS) (specific aim 2). In parallel, we will refine LCM methods with the goal of implementing them for future plant spaceflight genomics (specific aim 1). The latter aim will involve ground-based experiments in which specific steps in the LCM methods will be modified to mirror in-orbit operations. Upon successful completion of this proposal, we expect to generate new tools for conducting genomics in homogenous plant cell populations that will enable us to better differentiate microgravity effects from microgravity independent effects.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research seeks to fill technological gaps that will enable a better understanding of how plants develop in the microgravity environment of spaceflight. The focus of this research is to develop methods that will allow us to learn more about how expression of genes within specific root cell/tissue types change under microgravity or stress-induced spaceflight. The plant organ of interest in this proposal is the root. Because roots are crucial for uptake of water and nutrients to support optimal plant growth, results from this proposal will benefit agriculture on Earth. We expect that genes, which show cell/tissue type expression changes in response to gravity could be potential targets in pre-breeding efforts for developing crop cultivars that are resilient to environmental stress imposed by marginal lands on Earth and the confined growth environment during spaceflight.
Task Progress:	For this reporting period, we started work on experiments related to specific aim 1. At this stage of the project, we are currently in the process of optimizing methods to preserve root tissues for cell/tissue-type genomics. Our work thus far has involved testing the effects of different chemical fixatives, temperature, and duration on root RNA quality and integrity. A major result from our studies is that Farmer's fixative, which consists of ethanol and glacial acetic acid, proved to be the most promising for preserving root RNA. RNA extracted from roots fixed in Farmer's fixative for periods of up to four weeks at 4°C and -80°C had high RNA quality. In addition, roots stored in Farmer's fixative did not demonstrate shrinking or browning. While roots stored in RNAlater at 4°C produced higher RNA yield compared with Farmer's fixative, roots in this solution exhibited shrinking and browning indicating that root structure was compromised. Based on these results, roots fixed in RNAlater are not ideal for subsequent processing with LCM and will not be considered for further optimization. As an alternative approach to LCM, we plan to explore single-cell RNA sequencing to differentiate gravity responses from different cell populations in the root. This methodology involves protoplast generation, cell barcoding, and RNA sequencing. We have successfully completed protoplast optimization based on published protocols. Once optimization for both LCM and single cell RNA-Seq techniques are completed, four-day old maize seedlings will be gravistimulated by rotating their roots 90° against the Earth's gravity vector. 3-4 mm of root tips will be harvested and half of root samples will be processed using LCM technique, whereas the other half will be processed using the single-cell technique. We expect that direct comparisons of gene expression results from both techniques will guide us toward deciding the best future space omics approach. Development of bioinformatics tools to support NASA genomics work During this reporting period, we have also invested in developing bioinformatics tools that could help with analysis of whole genome sequencing data and biological material originating from our NASA-funded research. To this end, we have collaborated with the Noble Research Institute Enterprise Systems and Informatics (ESI) department. One bioinformatics software that has emerged from this collaboration is called TDNAScan. This software enables identification of causal mutations in transfer (T)-DNA mutagenized plants from next generation sequence (NGS) datasets. TDNAscan could provide a valuable addition to forward genetics toolkits because it enables causal gene identification particularly in mutants in which genes are disrupted by truncated T-DNA insertions. We believe that TDNA scan could be useful for future functional studies resulting from our LCM research.
Bibliography Type:	Description: (Last Updated: 01/13/2026)
Abstracts for Journals and Proceedings	De Bang L, Paez-Garcia A, Cannon AE, Chin S, Kolape J, Liao F, Sparks JA, Jiang Q, Blancaflor EB. "Brassinolide Inhibits Autotropic Root Straightening through Partial Disruption of the Actin Cytoskeleton." Presented at the Concurrent Technical Session: Plants 2. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019

Articles in Peer-reviewed Journals	De Bang L, Paez-Garcia A, Cannon AE, Chin S, Kolape J, Liao F, Sparks JA, Jiang Q, Blancaflor EB. "Brassinosteroids inhibit autotropic root straightening by modifying filamentous-actin organization and dynamics." Front Plant Sci. 2020 Feb 4;11:5. https://doi.org/10.3389/fpls.2020.00005 , Feb-2020
Articles in Peer-reviewed Journals	Sun L, Ge Y, Sparks JA, Robinson ZT, Cheng X, Wen J, Blancaflor EB. "TDNAscan: A software to identify complete and truncated T-DNA insertions." Front Genet. 2019 Jul 25;10:685. eCollection 2019. https://doi.org/10.3389/fgene.2019.00685 ; PMID: 31428129; PubMed Central PMCID: PMC6690219 , Jul-2019
Books/Book Chapters	Sadot E, Blancaflor EB. "The Actomyosin System in Plant Cell Division: Lessons Learned from Microscopy and Pharmacology." in "The Cytoskeleton: Diverse Roles in a Plant's Life." Ed. V. Sahi, F. Baluška. Cham : Springer, 2019. First Online 01 December 2019. Plant Cell Monographs, vol 24. https://doi.org/10.1007/978-3-030-33528-1_6 , Dec-2019